SOMAIYA VIDYAVIHAR UNIVERSITY

24.05.2025 (E)

Semester: Jan 2025 – April 2025		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: B.Tech. Computer Engineering	Class: TY	Semester:V(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 116U01E514	Name of the Course: Soft Computing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

PWD – Section B

Que. No.	Question	Max. Marks
Q4	Multiple choice questions	20
i)	In fuzzy logic, the truth value of a statement is: A. Always true or false B. Either 0 or 1 C. A value between 0 and 1 D. Only 0.5	1
ii)	Which of the following best describes the term "defuzzification" in fuzzy logic systems? A. Making a fuzzy set crisper B. Converting crisp input into fuzzy values C. Applying genetic mutation to fuzzy sets D. Removing fuzzy rules	1
iii)	Which activation function is commonly used in neural networks? A. Step function B. Sigmoid function C. ReLU (Rectified Linear Unit) D. All of the above	1
iv)	In fuzzy logic, a "membership function" defines: A. The number of fuzzy sets B. The rule base C. The degree to which an element belongs to a fuzzy set D. The input-output relationship .	1
v)	What is the primary function of an associative memory network? A. Error detection B. Memory compression C. Pattern recall based on input similarity D. Encryption and decryption	1
vi)	Hetero-associative memory maps: A. Input pattern to the same pattern	1

	B. Output pattern to the same input C. One set of patterns to another set D. Only binary data	
vii)	In bidirectional associative memory (BAM), which of the following is true? A. Only unidirectional recall is possible B. Learning is not required C. Recall can occur in both directions between input and output patterns D. It works only with continuous data	1
viii)	What type of learning does a Boltzmann machine typically use? A. Hebbian learning B. Backpropagation C. Simulated annealing D. Genetic algorithms	1
ix)	What type of learning does a Boltzmann machine typically use? A. Hebbian learning B. Backpropagation C. Simulated annealing D. Genetic algorithms	1
x)	A Neuro-Fuzzy System is a hybrid of: A. Genetic algorithms and neural networks B. Neural networks and fuzzy logic C. Decision trees and logic gates D. Clustering and regression	1

Que. No.	Question	Max. Marks
Q5	Solve **any two**	
i)	Explain the steps involved in designing a fuzzy rule-based control system. Each step :2M	10
ii)	Explain Kohonen Self Organizing Feature Map in detail with suitable diagrams. Also give its training algorithm. Network Architecture: 2M Explanation: 4M Training algorithm: 4M	10
iii)	Explain cognition neural network. 1. Introduction to cognitive neural networks and their purpose 2M 2. Explanation of the structure/architecture 2M 3. Description of layers and how information is processed 2M 4. Mention of biological inspiration / cognitive principles 1M 5. Real-world applications (e.g., pattern recognition, vision)1M 6. Clarity, terminology, and depth of explanation 2M	10

Que. No.	Question	Max. Marks
Q6	Solve **any two**	
i)	You are tasked with designing a **fuzzy logic controller** to maintain the **temperature** of a room by adjusting the **fan speed**. The controller must decide the fan speed based on the current **temperature error** (difference between desired and current temperature) and the **rate of change of error**	10

	(how fast the temperature is changing).
	Inputs:
	Error (E) = Desired Temp − Current Temp
	Change in Error (dE) = Current Error − Previous Error.
	Output:
	Fan Speed (FS)
	
	Tasks:
	1. **Define the fuzzy sets** for: [3M]
	  ○ Error (E): {Negative, Zero, Positive}
	  ○ Change in Error (dE): {Decreasing, Steady, Increasing}
	  ○ Fan Speed (FS): {Low, Medium, High}
	2. **Choose membership functions** (e.g., triangular or trapezoidal) for each fuzzy set. [1M]
	3. **Construct a fuzzy rule base** using if-then rules. Example: [2M]
	  ○ IF Error is Positive AND Change in Error is Increasing THEN Fan Speed is High
	4. **Describe the fuzzification, inference, and defuzzification process** used by the controller. [2M]
	5. **Simulate the fuzzy controller** (optional extension) for input values:. [2M]
	  ○ Error = 5°C, dE = -1°C, and determine the approximate fan speed using centroid method.	

ii)	Describe membership functions and their types. Definition and role of membership functions: 1M Three membership functions: each 3 M	10
iii)	Define fuzzy propositions and implications. Definition and explanation of fuzzy propositions: 4M Definition and explanation of fuzzy implications: 4M Mention of common implication operators or examples: 2M	10